



COMSATS UNIVERSITY ISLAMABAD
Lahore Campus
Department of Computer Science

☐ **Lab Mid Examination**

☒ **Terminal Examination Fall 2026**

Course Title:	DevOps for Cloud Computing				Course Code:	CSC418	Credit Hours:	3(2,1)
Course Instructor/s:	Dr. M. Hasanain Ch				Program Name	BS(CS)		
Semester:	6th	Batch:	FA23	Section:	A & B	Date:	18 May 2026	
Time Allowed:	90 Minutes				Maximum Marks:		50	
Student's Name:					Reg. No.			

Q1. [CLO: 5; Bloom Taxonomy Level <Applying>: Develop cloud native applications using current DevOps tools.]

General Instructions

1. You will continue with the **same project** used in the mid-term evaluation.
2. The project **must** now have:
 - a. A **Frontend**
 - b. A **Backend/API**
 - c. A **Database** (SQL or NoSQL)
3. A small or incomplete project will result in **mark deduction**.
4. All tasks must be documented clearly, with screenshots included in your final submission.

SECTION A: CONTAINERIZATION

(10 Marks)

Task A1: Docker Images

Create separate Dockerfiles for:

- Your **frontend application**
- Your **backend application**
- Your **database**

Task A2: Multi-Service Setup using Docker Compose

Prepare a docker-compose.yml that:

- Starts all three services
- Connects them in a common network
- Persists DB data

 Submit:

- Dockerfiles
- docker-compose.yml
- Screenshot of all containers running

SECTION B: CI/CD AUTOMATION

(10 Marks)

You may use **Jenkins**, **Azure DevOps**, or **GitHub Actions**.

Task B1: Pipeline Development

Build a pipeline that contains at least:

1. Build stage (frontend + backend)
2. Automated tests
3. Docker image build and push to registry
4. Deployment step to Kubernetes (or staging server)

Task B2: Trigger Configuration

Configure the pipeline so it runs on **push/commit** or **pull request**.

 Submit:

- Pipeline file (Jenkinsfile OR YAML workflow)
- Pipeline run screenshot (must show all stages completed)

SECTION C: KUBERNETES ON AZURE (AKS)

(10 Marks)

Deploy your 3-tier application on **Azure Kubernetes Service (AKS)**.

Task C1: Kubernetes Manifests

- Create an Azure Kubernetes Cluster (AKS).
- Deploy containerized application from Docker Hub onto AKS.
- Expose the app using a public IP address and provide a reachable link.

Task C2: AKS Deployment Verification

Your application must run correctly in the cluster.

Show:

- All pods in Running state
- Services created successfully
- Frontend connecting to backend
- Backend connecting to database

 Submit:

- Screenshots of kubectl get pods, kubectl get svc, and the running app.

SECTION D: SELENIUM AUTOMATED TESTING

(5 Marks)

Develop **Selenium test cases** for your application.

Task E1: Test Cases (Minimum 3)

Examples (choose any relevant to your project):

- Verify homepage loads
- Validate login or form behavior
- Check frontend-to-backend API response
- Validate navigation or button behavior

Task E2: Execution Report

Include the code and test execution screenshots.

 Submit:

- Selenium scripts
- Screenshot of test run

SECTION E: Report and Submission

(15 Marks)

Submit a short report containing:

- Viva will be conduct on **19 May 2026**
- Source code files
- Screenshots of output
- Assignment report in PDF or Word format
- All file must inside in folder and make zip file
- Assignment report (zip file) name should be your **Reg. NO**